COS 433 Problem Set 1

September 9, 2026

Due date: Wednesday, 9/23/2026 at 11:59pm See the course website for our submission and grading policies.

Proof citations. Your proofs may assume without proof any fact stated in lecture or stated on the problem sets. All other statements – including results from textbooks or other sources that were not mentioned in lecture – should be proved in your problem set write-up.

Grading Policy. A total score of 100 points constitutes full credit for the problem set. Each problem is worth 25 points. You are not expected to complete all of the problems. Feel free to work on whichever problems you prefer! **In our check-ins, you will only be asked about problems whose solutions you actually submitted.**

Problem 1: Shannon's Lower Bound (25 points)

Let $(\mathsf{Enc}, \mathsf{Dec})$ denote an encryption scheme for n-bit messages ($\mathcal{M} = \{0,1\}^n$) using λ-bit keys ($\mathcal{K} = \{0,1\}^\lambda$).

(a) Consider a randomized encryption scheme satisfying perfect security, but with a relaxed correctness property: for all messages m, it is guaranteed that

$$\Pr_{\mathsf{sk},r}\Big[\mathsf{Dec}(\mathsf{sk}, \mathsf{Enc}(\mathsf{sk}, m; r)) = m\Big] \geq 1 - \delta.$$

Prove that in this relaxed setting, for any constant $\delta > 0$, we still must have $\lambda \geq n - C_\delta$ for some constant C_δ (independent of n).

Optional: Suppose that $\delta = .1$. Can you show that $\lambda \geq n$?

(b) Consider a randomized encryption scheme satisfying perfect correctness, but with a relaxed *security property*. For any message m, let C_m denote the distribution

$$C_m = \Big\{\mathsf{sk} \leftarrow \{0,1\}^\lambda, r \leftarrow \{0,1\}^* : \mathsf{Enc}(\mathsf{sk}, m; r)\Big\}.$$

Our relaxed security property is that for all pairs of messages (m_0, m_1), the two distributions (C_{m_0}, C_{m_1}) have statistical distance at most ϵ.

Prove that in this relaxed setting, we still must have $\lambda \geq n - C_\epsilon$ for every constant $\epsilon > 0$.

Problem 2: Computational Indistinguishability (25 points)

(a) Let $X = \{X^{(\lambda)}\}_{\lambda \in \mathbb{N}}$ and $Y = \{Y^{(\lambda)}\}_{\lambda \in \mathbb{N}}$ be two ensembles of random variables. Suppose that $X \approx_c Y$.

Finally, let $A(x; r)$ be a polynomial-time randomized algorithm. Prove that the random variables $A(X; r)$ and $A(Y; r)$ (where r is uniformly random) are computationally indistinguishable.

(b) Let $\{(X_1^{(\lambda)}, X_2^{(\lambda)})\}_{\lambda \in \mathbb{N}}$ and $\{(Y_1^{(\lambda)}, Y_2^{(\lambda)})\}_{\lambda \in \mathbb{N}}$ denote two ensembles of *pairs* of random variables. We informally use (X_1, X_2) and (Y_1, Y_2) to denote these ensembles.

Suppose that $X_1 \approx_c Y_1$ and $X_2 \approx_c Y_2$. Is it the case that $(X_1, X_2) \approx_c (Y_1, Y_2)$? Prove or disprove.

Problem 3: PRG Transformations (25 points)

Throughout this problem, you may assume that PRGs exist.

Suppose that $G : \{0,1\}^\lambda \to \{0,1\}^n$ is a PRG with $n \geq 2\lambda$. Are the following constructions necessarily PRGs? Justify with a proof or a counterexample. (**Note:** make sure to prove that your counterexample G is really a PRG.)

(a) $G'(x) = G(G(x)_1 G(x)_2 ... G(x)_\lambda) \mid\mid G(G(x)_{\lambda+1} G(x)_{\lambda+2} ... G(x)_{2\lambda})$

(b) $G'(x) = G(x) \mid\mid G(\overline{x})$, where $\overline{x}$ is the complement of x

(c) $G'(x) = G(x) \mid\mid x_1 \oplus x_2 \ldots \oplus x_\lambda$

Problem 4: Attacking PRGs (25 points)

Show the following candidate function families G **do not** satisfy the definition of PRG.

(a) For all $\lambda \in \mathbb{N}$, let $G^{(\lambda)}$ be the following function mapping $\{0,1\}^{2\lambda}$ to $\{0,1\}^{2\lambda+1}$: On input (x, r) where $x, r \in \{0,1\}^\lambda$, G outputs $x \mid\mid r \mid\mid \langle x, r \rangle$ where $\langle x, r \rangle = \sum_{i=1}^{\lambda} x_i r_i \bmod 2$.

(b) For all $k \in \mathbb{N}$ and $\lambda = 2^k$ (otherwise, round λ up to the nearest power of 2), let $G^{(\lambda)}$ be the following function mapping $\{0,1\}^{\lambda+k}$ to $\{0,1\}^{\lambda+k+1}$:

- On input x, r where $x \in \{0,1\}^\lambda$ and $r \in \{0,1\}^k$, set $y = 0^{k+1} \mid\mid x$.
- Interpret r as the binary representation of a natural number $\bar{r}$ and define z to be string obtained by applying a "left circular shift" to y, $\bar{r}$ times. (A "left circular shift" of y is obtained by setting $y'_1 = y_2$, $y'_2 = y_3$, $\ldots$, $y'_{\lambda+k+1} = y_1$.)
- G outputs $y \oplus z$, the bit-wise XOR of strings y and z.

(c) For every $\lambda \in \mathbb{N}$, let $G^{(\lambda)}$ denote the following function mapping $\{0,1\}^{\lambda^2 \cdot n + \lambda \cdot n + \lambda} \to \{0,1\}^{\lambda^2 \cdot n + \lambda \cdot n + n}$:

$$G(A_1, \ldots, A_n, v_1, \ldots, v_n, s) = \Big(A_1, \ldots, A_n, v_1, \ldots, v_n, y\Big)$$

where

$$y = (s^\top A_1 s, s^\top A_2 s, \ldots, s^\top A_n s) \oplus (s^\top v_1, s^\top v_2, \ldots, s^\top v_n).$$

Here, each A_i is an $\lambda \times \lambda$ binary matrix, each v_i is a binary vector of length λ, and s is a binary vector of length λ.

Show that this choice of G is insecure whenever $n > \lambda^2 + \lambda$.

Problem 5: Distributing the Nuclear Codes (25 points)

The President is in possession of the launch codes of a nuclear weapon; the codes are represented as a **uniformly random binary string** $s \leftarrow \{0,1\}^\ell$ of length ℓ. The President has a cabinet of n Secretaries that she *mostly* trusts, but not *completely*.

In the event that the President is incapacitated or killed by a foreign adversary, she would like to ensure that her cabinet remains in possession of the nuclear codes. However, she does not want to give each Secretary a copy of the codes – too risky.

(a) Devise a scheme in which the President can ensure that:

- If all n Secretaries pool their information together, they can determine s.
- If *strictly fewer than* n of the Secretaries pool their information together, their probability of guessing s is $2^{-\ell}$.

Moreover, in your scheme, each Secretary should only need to store ℓ bits of information. Prove that your scheme satisfies all of these properties.

Hint in footnote below.[1]

(b) The President fears that the scheme from part (a) still risks losing the codes – what if the president *and* a secretary are not available?

Devise a scheme in which the President can ensure that:

- If *any two* of Secretaries pool their information together, they can determine s.
- *Any individual Secretary* can only guess s with probability $2^{-\ell}$.

In your scheme, each Secretary should only need to store $\le \ell \cdot n$ bits of information.

(c) Suppose that $n < 2^\ell$. Can you solve part (b) so that each Secretary only needs to store ℓ bits of information?

Hint in footnote below.[2]

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[1] Think about the one-time pad.

[2] One way to solve this problem involves finite fields (feel free to look them up).

Problem 6: More Computational Indistinguishability (25 points)

Recall the definition of computational indistinguishability from class: two distribution ensembles $X = \{X^{(\lambda)}\}, Y = \{Y^{(\lambda)}\}$ are computationally indistinguishable (denoted $X \approx_c Y$) if for all polynomial-time algorithms $D : \{0,1\}^* \to \{0,1\}$, the *distinguishing advantage*

$$\left|\Pr\left[D(X^{(\lambda)}) = 1\right] - \Pr\left[D(Y^{(\lambda)}) = 1\right]\right| = \epsilon(\lambda)$$

is a *negligible* function, meaning that for every constant C, $\epsilon(\lambda) < \lambda^{-C}$ for all but finitely many λ.

(a) Let $X_0, X_1, \ldots, X_t$ denote a finite collection of distribution ensembles, and suppose that $X_i \approx_c X_{i+1}$ for all i. Prove that $X_0 \approx_c X_t$.

(b) In cryptography, we often consider and compare an asymptotically growing number of distributions. Let $\{X_n\}_{n \in \mathbb{N}}$ be an infinite collection of distribution ensembles (meaning that for every n, $X_n = \{X_n^{(\lambda)}\}_\lambda$ as above). Define the "diagonal distribution" $X_\Delta = \{X_\lambda^{(\lambda)}\}_\lambda$, which represents the distribution "X_λ" asymptotically.

Suppose that for every natural number n, we have that $X_n \approx_c X_{n+1}$. Does it hold that $X_0 \approx_c X_\Delta$? Under what conditions can you show that $X_0 \approx_c X_\Delta$?

Problem 7: Length Extension (25 points)

Suppose that $\mathsf{SKE} = (\mathsf{Enc}, \mathsf{Dec})$ is a computationally secure one-time encryption scheme with keys of length λ and messages of length $\lambda + 1$.

For any constant C, show how to use this scheme to construct a computationally secure one-time encryption scheme SKE' with keys of length λ and messages of length $n(\lambda) = \lambda^C$. Prove that your construction satisfies all of the properties in the definition of an encryption scheme.

Problem 8: Do PRGs Exist? (25 points)

Prove that *there exists a function* $G = \{G^{(\lambda)} : \{0,1\}^\lambda \to \{0,1\}^{2\lambda}\}$ such that

$$G(U_\lambda) \approx_c U_{2\lambda}.$$

Does this imply that PRGs exist? Why or why not?

Hint in white font: